

Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Benchmark Clarifications:

Clarification 1: Instruction includes procedural fluency with the relationships of side lengths in special right triangles having angle measures of $30^\circ - 60^\circ - 90^\circ$ and $45^\circ - 45^\circ - 90^\circ$.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.1.2, MA.912.GR.1.6
- MA.912.GR.6.3

Terms from the K-12 Glossary

- Angle
- Equilateral Triangle
- Hypotenuse
- Isosceles Triangle
- Right Triangle

Vertical Alignment

Previous Benchmarks

- MA.8.GR.1.1, MA.8.GR.1.2

Next Benchmarks

- MA.912.T.1.5, MA.912.T.1.6, MA.912.T.1.7, MA.912.T.1.8
- MA.912.T.2
- MA.912.T.3

Purpose and Instructional Strategies

In grade 8, students solved problems involving right triangles using the Pythagorean Theorem. In Geometry, students use their understanding of sine, cosine and tangent to solve mathematical and real-world problems involving right triangles. In later courses, students will extend this knowledge to solve more difficult problems with right triangles, and extend the concept of trigonometric ratios to trigonometric functions on the unit circle and the number line.

- Within the Geometry course, the expectation is to use angle measures given in degrees and not in radians. Additionally, it is not the expectation for students to master the trigonometric ratios of secant, cosecant and cotangent within this course.
- It is customary to use Greek letters to represent angle measures (e.g., θ , α , β , γ).
- Problem types include cases where some information about the side lengths or angle measures of a right triangle is missing and one must use trigonometric ratios, inverse of trigonometric ratios or Pythagorean Theorem to determine the unknown length(s) or angle measure(s) within a mathematical or real-world context.
- Instruction includes the concept of inverse trigonometric ratios to determine unknown angle measures and how to find these values using technology, including a calculator. Students should have practice using both notations for the inverse trigonometric ratios ($\sin^{-1} A$ or $\arcsin A$; $\cos^{-1} A$ or $\arccos A$; and $\tan^{-1} A$ or $\arctan A$).
- Instruction includes exploring the relationships of the side lengths of special right triangles $45^\circ - 45^\circ - 90^\circ$ and $30^\circ - 60^\circ - 90^\circ$.
 - For example, students should realize that the special right triangle $45^\circ - 45^\circ - 90^\circ$ is an isosceles right triangle. Therefore, two of its angle measures and side

lengths are equivalent. So, if a side length is x units, then students can use the Pythagorean Theorem to determine that the hypotenuse is $x\sqrt{2}$ units.

Additionally, students can make the connection to its trigonometric ratios:

$\sin 45^\circ = \frac{1}{\sqrt{2}}$ (or equivalently $\frac{\sqrt{2}}{2}$); $\cos 45^\circ = \frac{1}{\sqrt{2}}$ (or equivalently $\frac{\sqrt{2}}{2}$); and $\tan 45^\circ = 1$.

- For example, students should realize that the special right triangle $30^\circ - 60^\circ - 90^\circ$ is half of an equilateral triangle. Students can use that knowledge to determine that the shorter leg is one-half the length of the hypotenuse. So, if the shorter leg is x units and the hypotenuse is $2x$ units, then students can use the Pythagorean Theorem to determine that the other leg is $x\sqrt{3}$ units. Additionally, students can make the connection to its trigonometric ratios such as, $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$ and $\tan 30^\circ = \frac{1}{\sqrt{3}}$ (or equivalently $\frac{\sqrt{3}}{3}$).

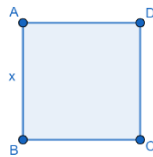
Common Misconceptions or Errors

- Students may choose the incorrect trigonometric ratio when solving problems.
- Students may misidentify the sides of triangles.
 - For example, students may identify the hypotenuse as being the adjacent leg or confuse the adjacent and opposite sides.

Instructional Tasks

Instructional Task 1 (MTR.3.1)

$ABCD$ is a square.



Part A. What is the measure of segment BD ?

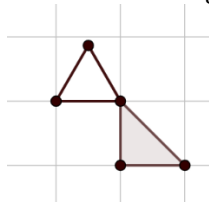
Part B. What is the measure of segment AC ?

Part C. If the measure of segment BD is 14 units, what is the measure of segment BC ?

Instructional Task 2 (MTR.7.1)

Part A. A company is requesting equilateral tiles to be made for their new office floor. If the height of the tile is approximately 10.4 inches, what is the length of the sides of the triangle?

Part B. The same company decides they also want to use half of a square with the side the same length as the height of the equilateral triangle. What is the length of the hypotenuse of the triangle formed from taking half of the square?



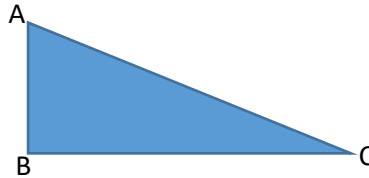
Instructional Items

Instructional Item 1

The logo of a local construction company contains an equilateral triangle. The height of the triangle is 10 units. What is the length of the measure of each side of the triangle?

Instructional Item 2

The right triangle ABC is shown. Angle B is the right angle and the length of AB is 1.5 centimeters and the length of BC is 3.1 centimeters.



Part A. Determine the measure of angles A and C .

Part B. Determine the length of AC .

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*